

Chapter

Absorption by Roots

1. During a practical exam, a plant cell in a particular solution was placed under a compound microscope. Students were told to observe the cell and name the tonicity of the solution and mention the process that occurred in the cell.

SQP 2025



- (a) Isotonic, Endosmosis
- (b) Hypotonic, Active Transport
- (c) Hypertonic, Endosmosis
- (d) Hypertonic, Exosmosis

2. Osmosis takes place when there is:

MAIN 2024

- (a) A freely permeable membrane
- (b) A cell wall
- (c) A selectively permeable membrane
- (d) An impermeable membrane

3. On a dry, sunny day, how does water vapour move through the stomata of a leaf? **SQP 2021**

- (a) Into the leaf by diffusion
- (b) Into the leaf by respiration
- (c) Out of the leaf by diffusion
- (d) Out of the leaf by respiration

4. Mineral salts are absorbed by roots from the soil in the form of: **SQP 2023**

- (a) Very dilute solution
- (b) Concentrated solution
- (c) Hypertonic solution
- (d) Very concentrated solution

5. Hydathode

MAIN 2022

- (a) Helps in transpiration
- (b) Helps in guttation
- (c) Helps in imbibition
- (d) Helps in transportation of water

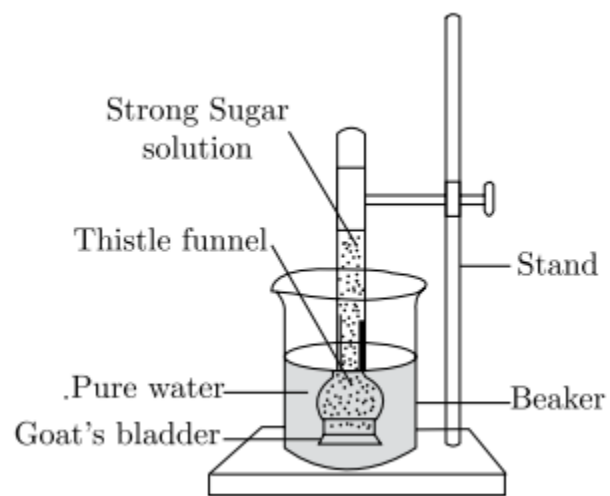
6. Phloem

SQP 2023

- (a) Helps in transportation of food
- (b) Helps in transportation of water
- (c) Helps in transportation of mineral
- (d) Helps in transportation of air

7. Expand the abbreviation-NADP. MAIN 2024
8. Unicellular outgrowths from the epidermis of mofs. MAIN 2024
9. One factor that affect water potential in plant. COMP 2023
10. Name the terms used in reference to concentration between two solutions.
SQP 2021
11. The tissue which conducts water and minerals from the root to the other parts of the plant. COMP 2021
12. A solution whose concentration is greater than the cell sap. COMP 2023
13. The condition of the cell when it is placed in a hypotonic solution.
MAIN 2021
14. The condition in which cell shrinks SQP 2023
15. The solution which has the same concentration of dissolved particles as inside the cell is called _____. MAIN 2023
16. The condition opposite to turgid is____. COMP 2021
17. Explain the term — Active transport SQP 2025
18. What is active transport in plants? COMP 2021
19. Draw a neat, labelled diagram of a root hair. SQP 2025
20. The apparatus shown was set up for an experiment. After some time the level of the liquid in the vertical section increased: SQP 2021
- (i) What process does the experiment demonstrate?
- (ii) Name two liquids that could be used in this experiment.
- (iii) Why did the level of the liquid in the vertical tube rise ?
- (iv) Explain how the process named in (i) affects a protozoan such as Amoeba.

(v) Give one example in plants in which the above process occurs and what is the advantage of this to plants.



Solution

1. (d) Hypertonic, Exosmosis

Water moves out of the cell in a hypertonic solution; this process is **exosmosis**.

2. (c) A selectively permeable membrane

Osmosis occurs when water moves across a **selectively permeable membrane** from a region of high water concentration to low water concentration.

3. (c) Out of the leaf by diffusion

On a dry, sunny day, water vapour **moves out of the leaf** through stomata by **diffusion**.

4. (a) Very dilute solution

Roots absorb mineral salts from the soil as a **very dilute solution** to facilitate easy uptake by the plant.

5. (b) Helps in guttation

Hydathodes are specialized structures at the leaf margins that release excess water in the form of **guttation**.

6. (a) Helps in transportation of food

Phloem transports **food (mainly sugars)** from leaves to other parts of the plant.

7. **NADP** – Nicotinamide Adenine Dinucleotide Phosphate

8. Root hairs

9. Number of moles of the substance present at atmospheric pressure.

10. Tonocity

11. Xylem

12. Hypertonic solution

13. Deplasmolysis

14. Flaccid

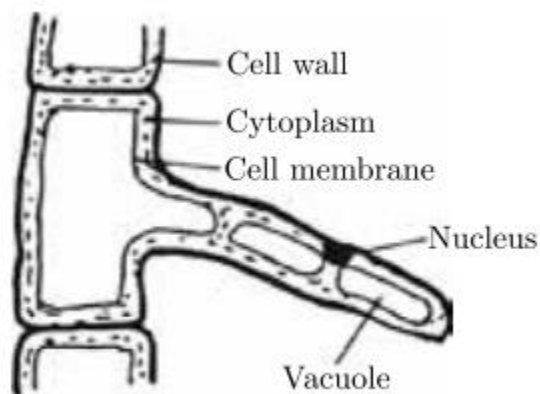
15. Isotonic

16. Flaccid

17. Active transport is the movement of **salts or ions** across a cell membrane from **lower to higher concentration** using **energy**.

18. Active transport in plants is the movement of **molecules from lower to higher concentration** across cell membranes, **requiring energy**.

19. Diagram of a root hair.



20. (i) Osmosis.

(ii) 1. Salt solution; 2. Water.

(iii) The vertical tube contains sugar solution and is separated from pure water by a semipermeable membrane. So, osmosis occurs and the level of the solution in the vertical tube rises.

(iv) Amoeba is a unicellular organism and lives in water. It contains plasma membrane. So, osmosis takes place and water enters into the cell. To prevent the bursting of the cell, Amoeba possesses contractile vacuole that throws out unwanted substances.

(v) Root hair – Osmosis helps in the absorption of water.